Week 8 – Tutorial

Tutorial Questions | Suggested duration: 50–60 minutes

SIS 3204Y(5) Natural Language Processing | Week 8 | Total: 30 marks

Instructions: Answer all questions. Marks are indicated alongside each part. Write clear, concise answers. For prompt-writing questions, state any rules needed to make the output unambiguous.

Part A – Fundamental Concepts (8 marks)

Question 1 – Foundation Models and LLMs (4 marks)

(a) Define a foundation model. (2 marks)

(b) Explain why not all foundation models are LLMs. Give one example. (2 marks)

Question 2 – How LLMs Work (4 marks)

(a) Explain the role of pre-training in developing an LLM. (2 marks)

(b) Explain two limitations of LLMs. (2 marks)

Part B – Prompt Engineering and In-Context Learning (10 marks)

Question 3 – Improving a Prompt (5 marks)

A university wants to classify student feedback as Positive, Negative or Neutral. The initial prompt is: “Analyse this feedback: The lecturer explains well, but the laboratory computers are frequently unavailable.”

(a) Identify two weaknesses of the initial prompt. (2 marks)

(b) Rewrite the prompt to make the task clearer and output more consistent. (3 marks)

Question 4 – In-Context Learning (5 marks)

(a) Differentiate between zero-shot and few-shot prompting. (2 marks)

(b) Construct a few-shot prompt for classifying customer messages as Order, Complaint or Enquiry. Include two demonstrations and one new message to classify. (3 marks)

Part C – LLM Evaluation and DeepEval (12 marks)

Question 5 – LLM Evaluation (4 marks)

(a) Explain why evaluating an LLM response requires more than checking whether it is relevant to the question. (2 marks)

(b) Differentiate between Answer Relevancy and Faithfulness. (2 marks)

Question 6 – Case Study: Evaluating a University Chatbot (8 marks)

A student asks: “What is the deadline for submitting the NLP assignment?” Retrieved university

document: “The NLP assignment must be submitted by 20 November at 16:00 through the university learning platform.”
Chatbot response: “The assignment deadline is 22 November at 16:00. You must submit it by email to your lecturer.”

(a) Identify two problems with the chatbot response. (2 marks)

(b) Explain how Answer Relevancy and Faithfulness would evaluate this response differently. (4 marks)

(c) Propose two measures to improve chatbot reliability. (2 marks)

Week 8 – Tutorial Model Answers & Marking Scheme

Tutorial answer key | 30 marks

SIS 3204Y(5) Natural Language Processing | Week 8 | Total: 30 marks

Award equivalent marks for technically correct alternative wording and well-justified examples. Do not require numerical DeepEval scores in Question 6.

Part A – Fundamental Concepts (8 marks)

Question 1 – Foundation Models and LLMs (4 marks)

(a) Define a foundation model. (2 marks)

Model answer: A foundation model is trained through broad, large-scale pre-training and can be adapted for multiple downstream tasks.

Mark allocation:

- Broad or large-scale pre-training (1)
- Adaptability to multiple downstream tasks (1)

(b) Explain why not all foundation models are LLMs. Give one example. (2 marks)

Model answer: Foundation models may focus on different modalities, such as images or audio, whereas LLMs primarily process and generate language. A vision foundation model is an example of a foundation model that is not necessarily an LLM.

Mark allocation:

- Correct distinction (1)
- Relevant example (1)

Question 2 – How LLMs Work (4 marks)

(a) Explain the role of pre-training in developing an LLM. (2 marks)
Model answer: Pre-training exposes the model to large amounts of text, enabling it to learn statistical patterns and language representations, often through next-token prediction.

Mark allocation:

- Learning from large-scale text data (1)
- Explains a training objective or learned patterns (1)

(b) Explain two limitations of LLMs. (2 marks)

Model answer: LLMs can hallucinate plausible but incorrect information and reproduce biases in their training data.

Mark allocation:

- One explained limitation (1)
- Second explained limitation (1)

Part B – Prompt Engineering and In-Context Learning (10 marks)

Question 3 – Improving a Prompt (5 marks)

A university wants to classify student feedback as Positive, Negative or Neutral. The initial prompt is: “Analyse this feedback: The lecturer explains well, but the laboratory computers are frequently unavailable.”

(a) Identify two weaknesses of the initial prompt. (2 marks)

Model answer: It does not state the allowed classification labels or explain how mixed sentiment should be classified.

Mark allocation:

- First valid weakness (1)
- Second valid weakness (1)

(b) Rewrite the prompt to make the task clearer and output more consistent. (3 marks)

Model answer: Classify the student feedback as exactly one of: Positive, Negative or Neutral. If both positive and negative views appear, use Neutral. Return only the label. Feedback: “The lecturer explains well, but the laboratory computers are frequently unavailable.” Expected output: Neutral.

Mark allocation:

- Task and labels clearly specified (1)
- Rule for mixed sentiment (1)
- Consistent output format (1)

Question 4 – In-Context Learning (5 marks)

(a) Differentiate between zero-shot and few-shot prompting. (2 marks)
Model answer: Zero-shot prompting provides instructions without task demonstrations; few-shot prompting includes labelled input-output examples in the prompt.

Mark allocation:

- Zero-shot correctly explained (1)
- Few-shot correctly explained (1)

(b) Construct a few-shot prompt for classifying customer messages as Order, Complaint or Enquiry. Include two demonstrations and one new message to classify. (3 marks)

Model answer: Classify each message as Order, Complaint or Enquiry.
Message: I would like to purchase two dresses. Label: Order
Message: My parcel arrived damaged. Label: Complaint
Message: Do you have this item in size XL? Label:
Expected output: Enquiry.

Mark allocation:

- Appropriate instruction and labels (1)
- Two correct demonstrations (1)
- New input and correct classification (1)

Part C – LLM Evaluation and DeepEval (12 marks)

Question 5 – LLM Evaluation (4 marks)

(a) Explain why evaluating an LLM response requires more than checking whether it is relevant to the question. (2 marks)

Model answer: A response can address the question yet contain incorrect or unsupported claims. Evaluation should also consider factual reliability and grounding.

Mark allocation:

- Relevance alone does not guarantee correctness (1)
- Need to assess accuracy or grounding (1)

(b) Differentiate between Answer Relevancy and Faithfulness. (2 marks)

Model answer: Answer Relevancy measures whether a response addresses the input question. Faithfulness assesses whether the response's claims are supported by supplied retrieval context.

Mark allocation:

- Answer Relevancy correctly explained (1)
- Faithfulness correctly explained (1)

Question 6 – Case Study: Evaluating a University Chatbot (8 marks)

A student asks: "What is the deadline for submitting the NLP assignment?" Retrieved university document: "The NLP assignment must be submitted by 20 November at 16:00 through the university learning platform."
Chatbot response: "The assignment deadline is 22 November at 16:00. You must submit it by email to your lecturer."

(a) Identify two problems with the chatbot response. (2 marks)

Model answer: It states 22 November instead of 20 November and says email instead of the university learning platform.

Mark allocation:

- Incorrect deadline identified (1)
- Incorrect submission channel identified (1)

(b) Explain how Answer Relevancy and Faithfulness would evaluate this response differently. (4 marks)

Model answer: Answer Relevancy: the chatbot addresses the question about the deadline, so it may be considered relevant despite being wrong. Faithfulness: the answer contradicts the retrieved document on the deadline and submission method; its claims are unsupported, so faithfulness should be poor.

Mark allocation:

- Defines Answer Relevancy (1)
- Applies relevance to the case (1)
- Defines Faithfulness (1)
- Identifies contradictions with context (1)

(c) Propose two measures to improve chatbot reliability. (2 marks)

Model answer: Require answers to be grounded in retrieved university information; test faithfulness and flag unsupported answers for review.

Mark allocation:

- First appropriate measure (1)
- Second appropriate measure (1)

Tutorial marks summary

Question	Topic	Marks
Q1	Foundation Models and LLMs	4
Q2	Pre-training and limitations	4
Q3	Prompt improvement	5
Q4	In-Context Learning	5
Q5	LLM evaluation	4
Q6	DeepEval chatbot case study	8
Total		30

Performance differentiation

🎬 Basic: recalls concepts but struggles to apply them.

🎬 Satisfactory: constructs workable prompts and explains limitations. 🎬 Good: uses ICL correctly and distinguishes evaluation dimensions. 🎬 Excellent: analyses relevant but unfaithful answers and proposes justified reliability controls.